Advanced Machine Learning, ETH Zurich

Hinge-fan networks

One hidden layer, one hinge per kink

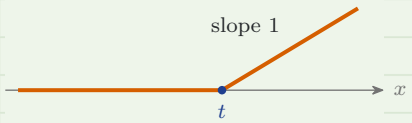
Carlos Cotrini

Autumn semester 2026

Entry 1

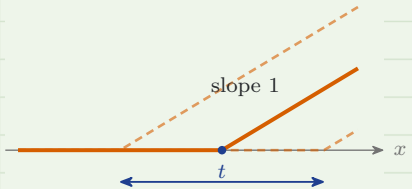
One layer of hinges

A hinge is the ReLU of a shifted input, $\mathrm{ReLU}(x - t)$: it stays at 0 left of t and rises with slope 1 right of it. Its position t is the only number it has.



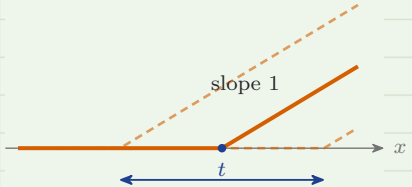
Solid: the hinge $\mathrm{ReLU}(x - t)$.

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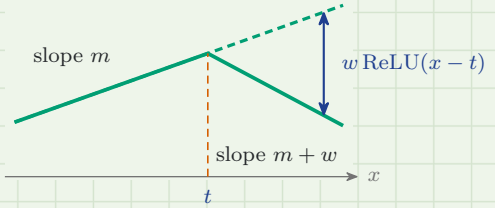


Solid: the hinge $\mathrm{ReLU}(x - t)$. Dashed: the same hinge at two other positions. Changing t slides it along x and keeps its shape.

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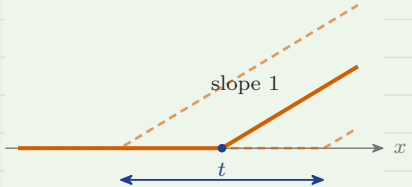


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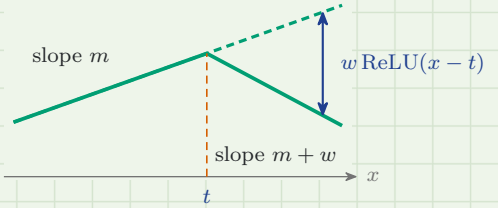


Add $w\,\mathrm{ReLU}(x - t)$ to a curve of slope m (dashed): from t on, its slope is $m + w$ (solid).

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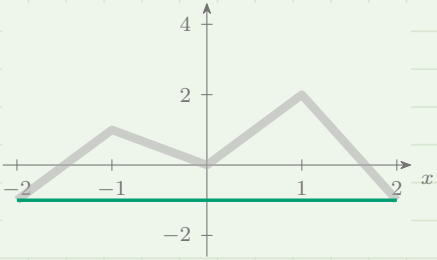


Add $w\,\mathrm{ReLU}(x - t)$ to a curve of slope m (dashed): from t on, its slope is $m + w$ (solid).

The weight w turns the curve at t, up for $w > 0$ and down for $w < 0$. In a network, t sits in the hinge and w on the arrow that leaves it.

Start from the constant c. A hinge at the left end a is a line on $[a, b]$, so it sets the first slope; every further hinge turns the curve at its position t_j by its weight w_j.

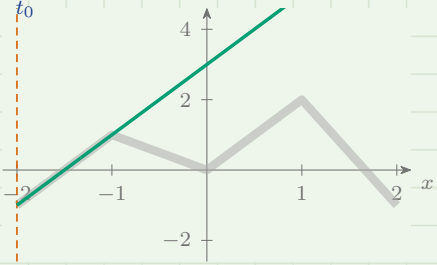
$$y = c + \sum_j w_j \,\mathrm{ReLU}(x - t_j)$$



Grey: the zigzag f. Green: the sum so far, one hinge more per click.

$y = -1$ the bias $c = f(-2)$

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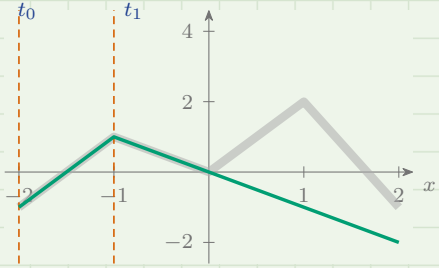
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$+\,2\,\mathrm{ReLU}(x+2)$ $t_0 = -2$, slope 0 to 2

On $[-2, 2]$, $\mathrm{ReLU}(x+2) = x+2$: hinge 0 sits at the left end and is a line, so with c it gives the first piece, $2x + 3$.

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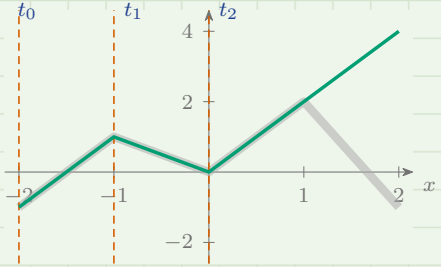
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$$\begin{aligned} y = {} & -1 && \text{the bias } c = f(-2) \\ & +2\,\mathrm{ReLU}(x+2) && t_0 = -2, \text{ slope 0 to 2} \\ & -3\,\mathrm{ReLU}(x+1) && t_1 = -1, \text{ slope 2 to } -1 \end{aligned}$$

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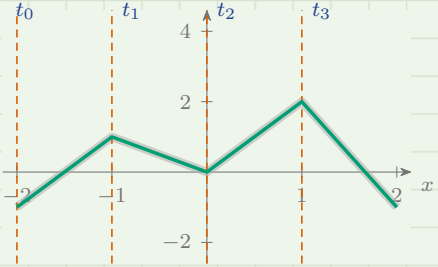
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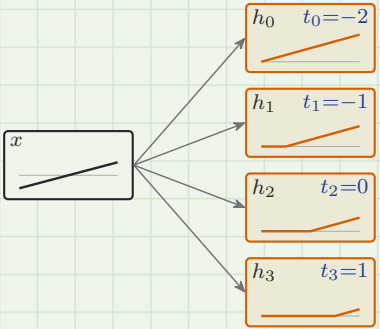
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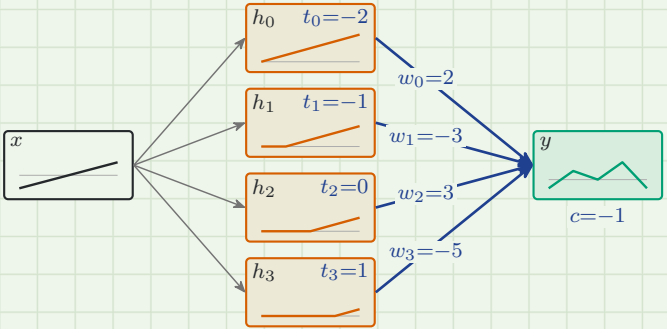
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Hinge j reads x with the fixed weight 1 and bends at its position t_j. The output y adds the hinges up, with the weights w_j and the bias c.



Every box draws its neuron on $[-2, 2]$. The hinges, on one scale, are one ramp slid to -2, -1, 0, 1.

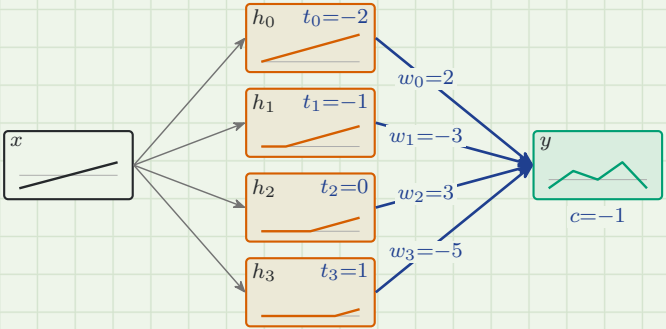
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$$h_j = \mathrm{ReLU}(x - t_j)$$

$$y = c + \textstyle\sum_j w_j h_j$$

Free, in pen: t_j and w_j for every hinge, and c: $2n + 1$ numbers, nine here. A grey arrow has the weight 1. The hinges can come in any order.

Entry 2

The construction

Read the numbers off f: the bias is its value at a, hinge 0 sits at a with the first slope m_0, and hinge i sits at the kink t_i with the change of slope $\Delta_i = m_i - m_{i-1}$ there.

output	$c = f(a)$		
hinge 0	$t_0 = a$,	$w_0 = m_0$	
hinge i	t_i, its kink,	$w_i = \Delta_i$	$i = 1, \dots, K$



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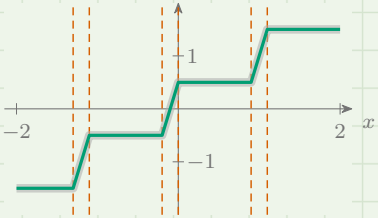
Then $y = f(a) + m_0 \,\mathrm{ReLU}(x - a) + \sum_i \Delta_i \,\mathrm{ReLU}(x - t_i)$,
and $y = f$: both are piecewise linear, they agree at a,
and on the ith piece both have the slope
$m_0 + \Delta_1 + \cdots + \Delta_i = m_i$.

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K kinks take $K + 1$ hinges, and K when f starts flat: then $m_0 = 0$ and hinge 0 goes.



The staircase of the page starts flat: six kinks, six hinges, $c = -1.5$ and $w_j = 5, -5, 5, -5, 5, -5$.

A hidden neuron usually has two numbers, $\mathrm{ReLU}(ux+v)$, and bends at $t=-v/u$. On $[a,b]$ each such neuron is a hinge of ours with a weight, plus at most a line and a constant.

$u>0$: $\mathrm{ReLU}(ux+v)=u\,\mathrm{ReLU}(x-t)$.
u moves onto the outgoing weight:
$\mathrm{ReLU}(\lambda z)=\lambda\,\mathrm{ReLU}(z)$ for $\lambda\geq 0$.

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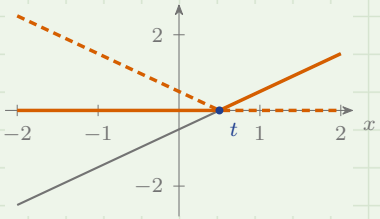
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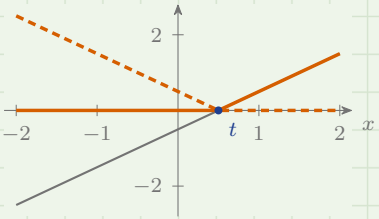
Dashed: $\mathrm{ReLU}(t-x)$. Solid: ours, $\mathrm{ReLU}(x-t)$. Grey: the line $x-t$, solid minus dashed.

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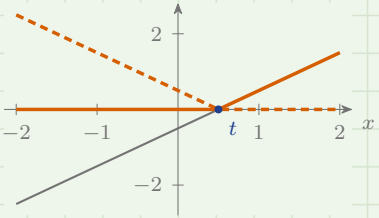
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So n arbitrary neurons make at most $n+1$ of our hinges: the one at a collects every line.



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Entry 3

Fitting

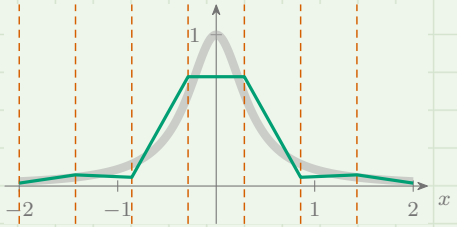
Fix the positions $t_1, \dots, t_n$. Then $y = c + \sum_j w_j \varphi_j(x)$, with the features $\varphi_j(x) = \mathrm{ReLU}(x - t_j)$, is linear in the numbers $\theta = (c, w_1, \dots, w_n)$ that are left.

On the points $x_1, \dots, x_N$, with f the target's values there, the mean squared error $\frac{1}{N}\|\Phi\theta - f\|^2$ is a convex quadratic in θ. Row i of the matrix Φ holds the features at x_i:

$$\big(1,\ \varphi_1(x_i),\ \dots,\ \varphi_n(x_i)\big).$$

Its minimizers solve the normal equations

$$\Phi^\top \Phi\, \theta = \Phi^\top f$$



Seven hinges spread evenly from -2 straddle the peak, and the best weights miss it: MSE 8.0×10^{-3}.

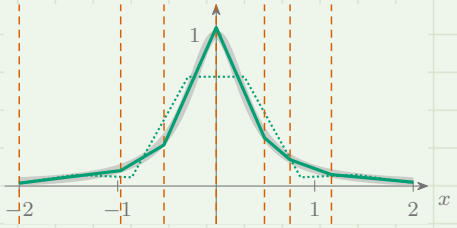
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Placed by the fit of the next frame, one hinge sits on the peak: MSE 1.3×10^{-4}, about sixty times lower. Dotted: the even fit.

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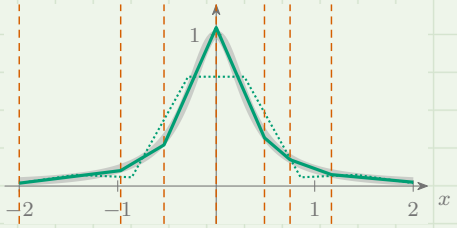
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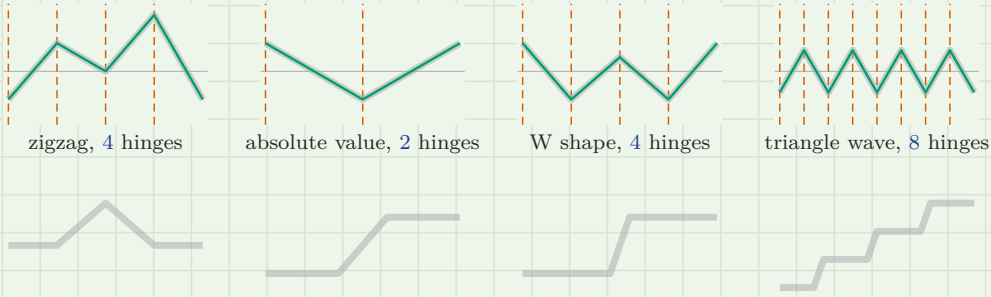
The hidden layer picks the features, and the output is linear regression on them, as in the course; random features pick the hidden layer at random. Placing the hinges is the nonconvex part.



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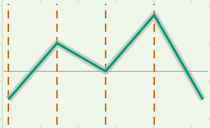
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The fit places the hinges by the dynamic program of the ladder page, with a hinge at a and without, sets c and the w_j by least squares, and keeps the smaller error.

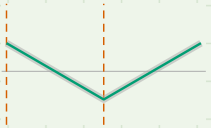


Grey: the target, under the fit; dashed: the hinges. Top row: one hinge per kink, and one for the first slope.

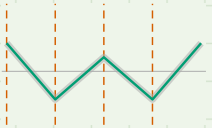
The fit places the hinges by the dynamic program of the ladder page, with a hinge at a and without, sets c and the w_j by least squares, and keeps the smaller error.



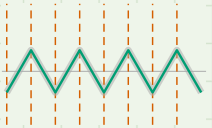
zigzag, 4 hinges



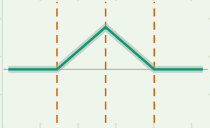
absolute value, 2 hinges



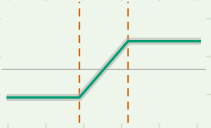
W shape, 4 hinges



triangle wave, 8 hinges



tent, 3 hinges



clipped ramp, 2 hinges



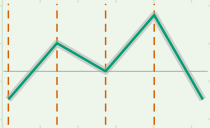
steep step, 2 hinges



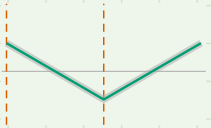
staircase, 6 hinges

Grey: the target, under the fit; dashed: the hinges. Top row: one hinge per kink, and one for the first slope. Bottom row: these start flat, so one per kink is enough.

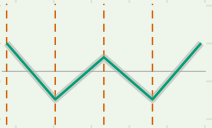
The fit places the hinges by the dynamic program of the ladder page, with a hinge at a and without, sets c and the w_j by least squares, and keeps the smaller error.



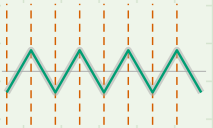
zigzag, 4 hinges



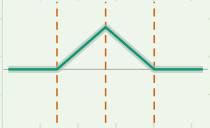
absolute value, 2 hinges



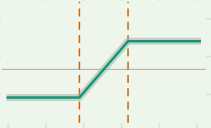
W shape, 4 hinges



triangle wave, 8 hinges



tent, 3 hinges



clipped ramp, 2 hinges



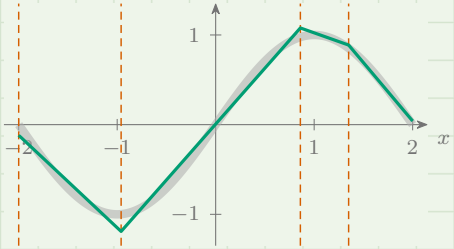
steep step, 2 hinges



staircase, 6 hinges

Grey: the target, under the fit; dashed: the hinges. Top row: one hinge per kink, and one for the first slope. Bottom row: these start flat, so one per kink is enough. No fit with one hinge fewer is exact.

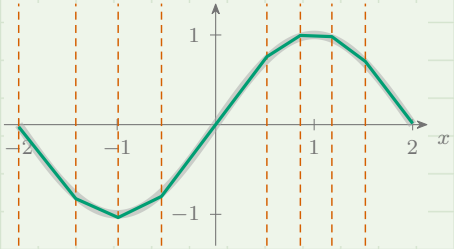
If $|f''| \le M$, the interpolant through $n+1$ evenly spaced points is a network of n hinges, and it stays within $2M/n^2$ of f on $[-2, 2]$.



The sine $\sin(\pi x/2)$ with 4 hinges: MSE 3.7×10^{-3}, largest error 0.19.

Each doubling of the hinges divides the largest error by about 4, and the MSE by about 16.

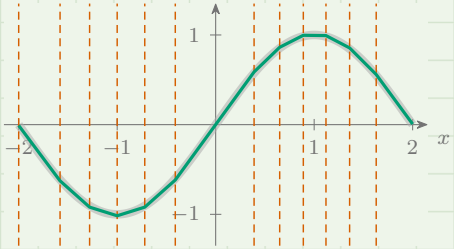
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The sine $\sin(\pi x/2)$ with 8 hinges: MSE 1.7×10^{-4}, largest error 0.040.

Each doubling of the hinges divides the largest error by about 4, and the MSE by about 16.

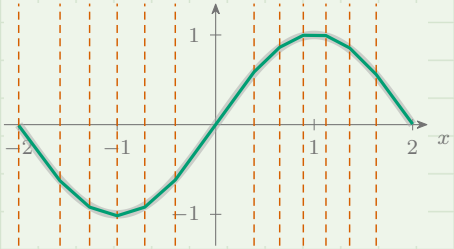
If $|f''| \le M$, the interpolant through $n+1$ evenly spaced points is a network of n hinges, and it stays within $2M/n^2$ of f on $[-2, 2]$.



The sine $\sin(\pi x/2)$ with 12 hinges: MSE 3.3×10^{-5}, largest error 0.017.

Each doubling of the hinges divides the largest error by about 4, and the MSE by about 16.

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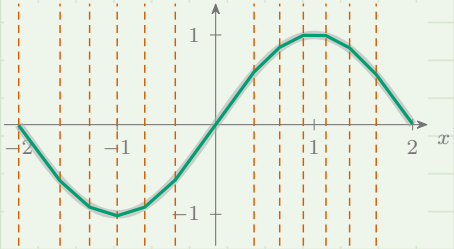
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Hinges until the largest error drops below 0.02, about two pixels on the page:

parabola	9	sine	11
Runge function	9	cubic	11
Gaussian bump	11	two periods	21
		chirp	24

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The ladder makes the same functions with as many layers; its page interpolates f at the kinks. With $n = 10$, least squares lowers the MSE about 4 to 6 times: for the sine, 7.0×10^{-5} against 3.6×10^{-4}.

Entry 4

Summary

The hinge chain, the hinge ladder and the hinge fan all build a piecewise-linear function out of hinges. They differ in depth, in width, and in what each hinge reads.

	Hinge chain	**Hinge ladder**	**Hinge fan**
Hidden layers	K	$K+1$	1
Neurons per layer	2	3	$K+1$ (K if f starts flat)
A hinge reads	the hinge before it	the copy of x	x

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ReLU on	every neuron, lift C	the hinges	the hinges
Free numbers	$\alpha_\ell, \ldots, \varepsilon_\ell$; $\omega_h, \omega_g, \omega_0$	w_ℓ, b_ℓ; b_0	t_j, w_j; c
The kinks	in order, by the gaps	anywhere	anywhere

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The fan is the ladder folded flat: the ladder's running sum adds one hinge per layer, and the fan's output adds them all at once. The ladder's $y = x + b_0 + \sum_\ell w_\ell \,\mathrm{ReLU}(x + b_\ell)$ is a fan whose hinge at a carries the x.

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On the page *Hinge-fan networks*, pick a target, add or remove hinges, move every t_j, w_j and c by hand, or let the fit place the hinges and solve for the weights.